Distinguishing Load Torque Oscillations and Eccentricity Faults in Induction Motors Using Stator Current Wigner Distributions

Martin Blödt, Jérémi Regnier, and Jean Faucher, *Member, IEEE*

***Abstract*—This paper proposes a novel diagnosis method for detection and discrimination of two typical mechanical failures in induction motors by stator current analysis: load torque oscillations and dynamic rotor eccentricity. A theoretical analysis shows that each fault modulates the stator current in a different way: torque oscillations lead to stator current phase modulation, whereas rotor eccentricities produce stator current amplitude modulation. The use of traditional current spectrum analysis involves identical frequency signatures with the two fault types. A time–frequency analysis of the stator current with the Wigner distribution leads to different fault signatures that can be used for a more accurate diagnosis. The theoretical considerations and the proposed diagnosis techniques are validated on experimental signals.**

***Index Terms*—Condition monitoring, eccentricity, fault diagnosis, induction motor drives, load unbalance, motor current signature analysis, torque oscillation, Wigner Distribution (WD).**

I. Introduction

In order to increase the productivity, reliability, and safety of an installation containing induction motors, monitoring techniques have been more and more investigated in industrial applications. Instead of a cost-expensive vibration analysis for mechanical fault detection, stator current based condition monitoring can be advantageous because the current signal is often already available or relatively inexpensive to measure.

Traditional detection techniques for mechanical faults, such as shaft misalignment, eccentricity, or load unbalance, monitor frequency combinations in the stator current spectrum [1]–[3]. The relevant frequencies are $f_s \pm f_r$, where f_s is the supply frequency and f_r the shaft rotational frequency.

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M. Blödt is with the Mobility Division, Siemens AG, 91052 Erlangen, Germany (e-mail: blodt@laplace.univ-tlse.fr; martin.bloedt@siemens.com).

J. Regnier and J. Faucher are with the Laboratoire Laplace, 31071 Toulouse, France (e-mail: regnier@laplace.univ-tlse.fr; jean.faucher@laplace.univ-tlse.fr).

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In this paper, mechanical failures are classified into faults that cause air gap eccentricity, load torque oscillations, or a combination of both. Load torque oscillations can be caused by the following:

1) load unbalance (not necessarily a fault but can also be inherent to the load type);
2) shaft misalignment;
3) gearbox fault, e.g., broken tooth;
4) bearing faults.

Air gap eccentricity, i.e., a nonuniform air gap, can be, for example, the consequence of bearing wear or bearing failure, bad motor assembly with rotor unbalance, or a rotor which is not perfectly centered. In general, eccentricity is a sign for a mechanical problem within the electric motor, whereas load torque oscillations point to a fault that is located outside of the motor.

Previous works have shown that load torque oscillations lead to a particular stator current modulation which is a sinusoidal phase modulation (PM) [4]. In contrast, it will be shown in the following that air gap eccentricity leads to stator current amplitude modulation (AM). The classical current spectrum based techniques cannot distinguish between AM and PM when modulation indexes are small. Thus, dynamic eccentricity cannot be discriminated from load torque oscillations at f_r. This paper proposes an approach to overcome this problem based on stator current time–frequency analysis.

Section II presents the effect of load torque oscillations and dynamic eccentricity on the stator current. Two resulting fault models with sinusoidal PM and AM at shaft rotational frequency are established. Moreover, load torque oscillations generated by eccentricity are discussed. In Section III, the chosen signal-processing method i.e., the Wigner distribution (WD), is described, and the theoretical fault signatures are calculated for PM and AM signals. Section IV presents the experimental setup used to validate the theoretical fault signatures. A suitable procedure for extraction of a fault indicator is described and illustrated with experimental results.

II. Stator Current Signal Models

A. Load Torque Oscillations

The following section on load torque oscillations has been detailed in [4]. Mechanical faults can lead to a periodically varying load torque at characteristic fault frequency f_c.

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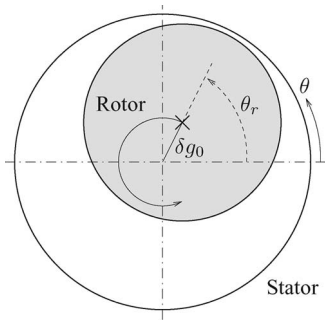


Fig. 1. Schematic representation of dynamic eccentricity.

Considering only the fundamental component in the Fourier series development, the load torque can be described by

$$\Gamma_{\rm load}(t) = \Gamma_{\rm const} + \Gamma_c \cos(\omega_c t) \tag{1}$$

where Γ_c is the amplitude of the load torque oscillation and $\omega_c = 2\pi f_c$. The mechanical speed $\omega_r(t)$ is obtained by solving the mechanical equation and contains an oscillating term resulting from the torque oscillation. Consequently, the mechanical rotor position $\theta_r(t)$ includes an oscillating term as well:

$$\theta_r(t) = \frac{\Gamma_c}{J\omega_c^2}\cos(\omega_c t) + \omega_{r0} t \tag{2}$$

where J denotes the total inertia and ω_{r0} is the average mechanical speed. These oscillations lead to a PM of the rotor magnetic field expressed in the stator reference frame, whereas the stator magnetic field is not directly affected.

In the presence of load torque oscillations, the induction motor stator current can be considered as the sum of a phase modulated sinusoidal signal $i_{rt}(t)$ and an unmodulated signal $i_{st}(t)$ [4],

$$\begin{aligned} i_{\rm pm}(t) &= i_{st}(t) + i_{rt}(t) \\ &= I_{st}\sin(\omega_s t + \varphi_s) + I_{rt}\sin\left[\omega_s t + \beta\cos(\omega_c t)\right] \end{aligned} \tag{3}$$

where $i_{st}(t)$ and $i_{rt}(t)$ denote the stator current components resulting from the stator and rotor magnetic field with amplitudes I_{st} and I_{rt}, respectively. β is the PM index proportional to Γ_c/ω_c^2. A healthy case is obtained for $\beta = 0$. Note that in this simplified model, the stator current component is not affected by the PM, i.e., the secondary and higher order armature reactions are neglected.

B. Air Gap Eccentricity

Air gap eccentricity leads to an air gap that is no longer constant with respect to the circumference angle θ and/or time. In the case of dynamic eccentricity, the rotor rotational center is different from its geometrical center (X) but identical to the geometrical stator center (see Fig. 1). The air gap length varies as a function of the circumference angle θ and the mechanical rotor angle $\theta_r = \omega_r t$. The point of minimum air gap length is moving, and it is not stationary in contrast to static eccentricity [5].

The air gap length $g_{de}(\theta, t)$ can be approximated for a small air gap and low levels of dynamic eccentricity by the following expression [3]:

$$g_{de}(\theta, t) \approx g_0\left(1 - \delta_d \cos(\theta - \omega_r t)\right) \tag{4}$$

where δ_d denotes the relative degree of dynamic eccentricity and g_0 is the mean air gap length without eccentricity. Note that in the case of static eccentricity, $g_{se}(\theta, t)$ takes the same expression but with $\omega_r = 0$, i.e., the point of minimum air gap length is stationary

$$g_{se}(\theta, t) \approx g_0\left(1 - \delta_s \cos(\theta)\right) \tag{5}$$

with δ_s the relative degree of static eccentricity. Since dynamic eccentricity is more general, it will mainly be considered in the following.

The air gap permeance $\Lambda(\theta, t)$ is obtained as the inverse of $g(\theta, t)$ multiplied by the permeability of free space μ_0. Following a classical approach, the permeance is written as a Fourier series [5]

$$\Lambda_{de}(\theta, t) = \frac{\Lambda_0}{2} + \sum_{i_{\rm ecc}=1}^{\infty} \Lambda_{i_{\rm ecc}} \cos(i_{\rm ecc}\theta - i_{\rm ecc}\omega_r t) \tag{6}$$

where $\Lambda_0/2 = \mu_0/g_0$ is the permeance without eccentricity. The higher order coefficients of the Fourier series can be written as [5]

$$\Lambda_{i_{\rm ecc}} = \frac{2\mu_0(1 - \sqrt{1-\delta^2})^{i_{\rm ecc}}}{g_0 \delta^{i_{\rm ecc}}\sqrt{1-\delta^2}}. \tag{7}$$

Dorrell has shown in [6] that the coefficients with $i_{\rm ecc} \geq 2$ are rather small for $\delta < 40\%$. For the sake of simplicity, they are neglected in the following considerations.

The air gap flux density is the product of permeance with the magnetomotive force (MMF) [7]. The total fundamental MMF wave can be written as

$$F_{\rm tot}(\theta, t) = F_1 \cos(p\theta - \omega_s t - \varphi_t) \tag{8}$$

with φ_t being the initial phase. Hence, the flux density in the presence of dynamic eccentricity is

$$B_{de}(\theta, t) \approx B_1\left[1 + 2\frac{\Lambda_1}{\Lambda_0}\cos(\theta - \omega_r t)\right]\cos(p\theta - \omega_s t - \varphi_t) \tag{9}$$

with $B_1 = \Lambda_0 F_1$.

The fraction $2\Lambda_1/\Lambda_0$ is equal approximately to δ_d for small levels of eccentricity. The air gap flux density can therefore be written as

$$B_{de}(\theta, t) = B_1\left[1 + \delta_d \cos(\theta - \omega_r t)\right]\cos(p\theta - \omega_s t - \varphi_t). \tag{10}$$

This equation shows the fundamental effect of dynamic eccentricity on the air gap magnetic flux density; the modified air gap permeance causes an AM of the fundamental flux-density wave with respect to time and space. The AM index is approximately the degree of dynamic eccentricity δ_d.

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In the case of static eccentricity, the fundamental flux density is expressed as

$$B_{se}(\theta,t) = B_1\left[1+\delta_s\cos(\theta)\right]\cos(p\theta-\omega_s t-\varphi_t) \qquad (11)$$

which shows that static eccentricity leads only to flux density AM with respect to space.

Consequently, the AM can also be found on the stator current $i_{\mathrm{am}}(t)$ that is expressed as follows, in the case of dynamic eccentricity:

$$i_{\mathrm{am}}(t) = I_1\left[1+\alpha\cos(\omega_r t)\right]\cos(\omega_s t-\varphi_i). \qquad (12)$$

In this expression, I_1 denotes the amplitude of the stator current fundamental component, and α denotes the AM index which is proportional to the degree of dynamic eccentricity δ_d. Static eccentricity does not lead to frequencies different from ω_s, since the corresponding additional flux-density waves are also at the supply pulsation.

C. Torque Oscillations Resulting From Eccentricity

This section explains theoretically through the MMF and permeance-wave approach how eccentricity leads to an increase of oscillating-torque components at f_r. This has been observed through numerical simulations of induction machines with static eccentricity in [8] and both types of eccentricity in [9]. However, a theoretical demonstration in literature is not known to the author.

According to the considerations in [7], the motor output torque Γ is produced by the interaction of stator and rotor flux-density waves, denoted B_s and B_r, respectively, with the derivatives of the rotor flux-density waves

$$\Gamma = -\frac{1}{\mu_0} l_m R_0 g_0 \int_0^{2\pi}\left[B_s(\theta,t)+B_r(\theta,t)\right]\frac{\partial}{\partial\theta_r}B_r(\theta,t)\,d\theta \qquad (13)$$

where θ_r denotes the rotor position, l_m denotes the effective machine length, and R_0 is the mean air gap radius. The interacting waves must have equal pole-pair numbers or if not, the resulting torque is zero.

The preceding section [see (4)–(10)] has shown that the modified air gap length with dynamic eccentricity modulates the amplitude of the flux-density waves. In (10), the total MMF, being the sum of rotor and stator MMF, was considered. For the following development, stator and rotor MMF (and also flux-density waves) will be regarded separately. Supposing dynamic eccentricity, the rotor and stator flux-density waves can be rewritten as

$$\begin{aligned} B_s(\theta,t) &= B_{s1}\cos(p\theta-\omega_s t-\varphi_s) \\ &\quad + B_{s1}\delta\cos\left[(p\pm1)\theta-(\omega_s\pm\omega_r)t-\varphi_s\right] \qquad (14)\\ B_r(\theta,t) &= B_{r1}\cos(p\theta-\omega_s t-\varphi_r) \\ &\quad + B_{r1}\delta\cos\left[(p\pm1)\theta-(\omega_s\pm\omega_r)t-\varphi_r\right]. \qquad (15)\end{aligned}$$

B_{s1} and B_{r1} denote the amplitude of the fundamental rotor or stator flux-density wave, and φ_s and φ_r denote the respective phase angles. It becomes clear through these expressions that with respect to the uniform air gap, dynamic eccentricity leads to additional flux-density waves with $p\pm1$ pole pairs and pulsation $\omega_s\pm\omega_r$. It will be supposed in the following that $p>1$ in order to avoid homopolar flux.

The derivatives $\partial B_r(\theta,t)/\partial\theta_r$ are waves with the same pole-pair number and pulsation as $B_r(\theta,t)$; only the amplitude and phase change. Therefore, the corresponding integrals take the following general form without constant factors:

$$\begin{aligned} I(t) &= \int_0^{2\pi}\cos\left(k_1\theta-\varphi_1(t)\right)\cos\left(k_2\theta-\varphi_2(t)\right)d\theta \\ &= \frac{1}{2}\int_0^{2\pi}\cos\left[(k_1+k_2)\theta-(\varphi_1(t)+\varphi_2(t))\right] \\ &\quad +\cos\left[(k_1-k_2)\theta-(\varphi_1(t)-\varphi_2(t))\right]d\theta \qquad (16)\end{aligned}$$

with $k_1,\ k_2\in\mathbb{N}$. The first term of this integral is always zero. The second term can only be different from zero if $k_1=k_2$, i.e., a nonzero torque is only produced if the pole-pair numbers of the interacting waves are equal.

If the interacting flux-density waves have the same pulsation ω_s such as the stator and rotor fundamental, they produce constant torque with respect to time, since $\varphi_1(t)-\varphi_2(t)$ will not be a function of time in this case. Time-varying torque is produced if the waves have equal pole-pair numbers and different pulsations.

Pure static eccentricity leads to additional flux-density waves with pole-pair numbers $p\pm1$ and pulsation ω_s [see (11)]. Therefore, no time-varying torque is produced, since the pulsation is always ω_s. Pure dynamic eccentricity results in additional flux-density waves with pole-pair numbers $p\pm1$ and pulsation $\omega_s\pm\omega_r$ [see (10)], i.e., the waves with the same pole-pair number also have the same pulsation. Therefore, no time-varying torque can be produced either. Hence, pure dynamic or pure static eccentricity does not lead to waves with equal pole-pair numbers and different pulsations if higher order armature reactions are neglected.

However, realistic eccentricity faults in practice include a mixture of static and dynamic eccentricity simultaneously, which is termed mixed eccentricity. In this case, additional $p\pm1$ pole-pair waves with pulsations ω_s and $\omega_s\pm\omega_r$ are present. The interaction of these waves produces oscillating-torque components at shaft rotational frequency ω_r. Consider the following example with stator-related flux-density waves $B_{s,se}(\theta,t)$ reflecting static eccentricity:

$$B_{s,se}(\theta,t) = \delta_s B_s\cos\left[(p\pm1)\theta-\omega_s t\right]. \qquad (17)$$

The derivatives of the rotor flux-density waves resulting from dynamic eccentricity are of the following form:

$$\frac{\partial}{\partial\theta_r}B_{r,de}(\theta,t) = \delta_d B_r'\cos\left[(p\pm1)\theta-(\omega_s\pm\omega_r)t-\varphi_r\right]. \qquad (18)$$

The associated torque components can be calculated from (13). Considering only the particular interaction between

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$B_{s,se}(\theta,t)$ and $\partial B_{r,de}(\theta,t)/\partial\theta_r$, the oscillating-torque components of the following form are obtained:

$$\Gamma_{se,de}(t) = -\frac{\pi l_m R_0 g_0}{\mu_0}\delta_s\delta_d B_s B_r' \sin(\mp\omega_r t - \varphi_r). \quad (19)$$

Other interactions in the case of mixed eccentricity are also capable of producing oscillating torques at ω_r, e.g., stator-related waves resulting from dynamic eccentricity and rotor-related waves resulting from static eccentricity.

Until now, secondary and higher order armature reactions i.e., the reaction of the stator currents on the rotor fields, have not been considered. However, they may also explain torque oscillations in consequence of pure dynamic eccentricity. According to (12), dynamic eccentricity leads to stator current components at $f_s \pm f_r$. Consequently, these currents lead to stator fields with p pole pairs and pulsations $\omega_s \pm \omega_r$. Interactions between the latter and the rotor fundamental flux density produce also oscillating torque components at ω_r.

The preceding theoretical considerations have an important consequence for the detection of eccentricity faults by stator current monitoring. It has been demonstrated that mixed eccentricity leads to oscillating torques at ω_r. The same result is obtained with pure dynamic eccentricity if higher order armature reactions are considered. The eccentricity-related torque oscillations lead also to a PM of the stator current following the mechanism described in Section II-A. It can therefore be concluded that eccentricity faults lead to both stator current AM and PM at the same modulation frequency f_r.

III. Wigner Distribution of Stator Current Models

In steady-state conditions, the commonly used signal-processing method for current-based condition monitoring is the power-spectral analysis. It will be shown in the following that this method is insufficient for distinguishing PM and AM.

One possible approach to overcome this problem is the use of time–frequency signal analysis. These methods can be used in the case of variable-speed drives [10], but they can also provide valuable information for diagnosis during operation at constant speed. The proposed time–frequency method is the Wigner Distribution (WD), defined as [11]

$$W_x(t,f) = \int_{-\infty}^{+\infty} x\left(t+\frac{\tau}{2}\right) x^*\left(t-\frac{\tau}{2}\right) e^{-j2\pi f\tau} d\tau \quad (20)$$

where x^* denotes the complex conjugate of x. The WD can be seen as the Fourier transform (FT) of a kernel $K_x(t,\tau) = x(t+\tau/2)x^*(t-\tau/2)$ with respect to the delay τ.

In order to simplify the theoretical calculations, all signals will be considered in their complex, analytical form [11]. The real signal is obtained as the real part of the analytical signal.

A. Spectrum of PM and AM Signals

The power-spectral density (PSD) $P_x(f)$ of a discrete signal $x[n]$ is usually estimated using the periodogram estimator which is defined as [12]

$$P_x(f) = \frac{1}{N}\left|\mathrm{FT}\left\{x[n]\right\}\right|^2 \quad (21)$$

where FT denotes the FT and N is the sample length.

The FT of a PM signal is well known from communication theory [13]. The FT absolute value of the analytical signal corresponding to (3) can be expressed as follows:

$$|I_{\mathrm{pm}}(f)| = I_{st}\delta(f-f_s) + I_{rt}\sum_{n=-\infty}^{+\infty} J_n(\beta)\delta\left(f-(f_s+nf_c)\right) \quad (22)$$

where J_n denotes the nth-order Bessel function of the first kind and $\delta(f)$ is the Dirac delta function. For small modulation indexes β, the Bessel functions of the order $n \geq 2$ are very small and may be neglected (narrow-band approximation).

Using this approximation, $|I_{\mathrm{pm}}(f)|$ is expressed as

$$|I_{\mathrm{pm}}(f)| = \left(I_{st} + I_{rt}J_0(\beta)\right)\delta(f-f_s) + I_{rt}J_1(\beta)\delta\left(f-(f_s\pm f_c)\right). \quad (23)$$

It becomes clear through this expression that the fault leads to sideband components of the fundamental at $f_s \pm f_c$.

In the case of the AM signal (12), the FT modulus of the corresponding analytical signal is

$$|I_{\mathrm{am}}(f)| = I_1\delta(f-f_s) + \frac{1}{2}\alpha I_1\delta\left(f-(f_s\pm f_r)\right) \quad (24)$$

Therefore, an AM stator current signal with sinusoidal AM also involves sidebands at $f_s \pm f_r$. This signifies that a distinction of the two types of modulation is not possible by power-spectral analysis when the torque oscillations are at $f_c = f_r$.

B. WD of PM Signal

The stator current in the presence of load torque oscillations can be considered as the sum of a pure frequency and a phase-modulated signal [see (3)]. The detailed calculations of the stator current WD can be found in [4]. The following approximate expression is obtained for small β:

$$W_{i_{\mathrm{pm}}}(t,f) \approx \left(I_{rt}^2 + I_{st}^2\right)\delta(f-f_s) - I_{rt}^2\beta\sin(\omega_c t)\,\delta\left(f-f_s-\frac{f_c}{2}\right) + I_{rt}^2\beta\sin(\omega_c t)\,\delta\left(f-f_s+\frac{f_c}{2}\right). \quad (25)$$

The WD of the PM stator current is therefore a central frequency at f_s with sidebands at $f_s \pm f_c/2$. These components have time-varying amplitudes at frequency f_c. It is important to note that the lower sideband has the opposite sign to the upper sideband for a given point in time, i.e., a phase shift of π exist theoretically between the two sidebands.

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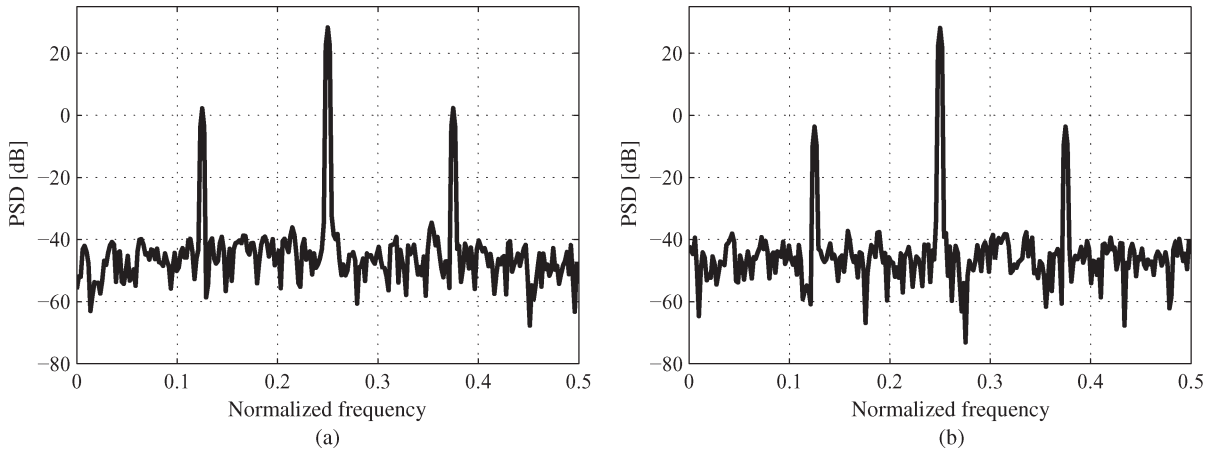


Fig. 2. Power-spectral density of synthesized (a) PM and (b) AM signals.

C. WD of AM Signal

In the following, the WD of a stationary AM signal according to (12) is calculated. The kernel $K_{i_{\rm am}}(t,\tau)$ is expressed as

$$\begin{aligned} K_{i_{\rm am}}(t,\tau) &= i_{\rm am}\left(t+\frac{\tau}{2}\right) i_{\rm am}^*\left(t-\frac{\tau}{2}\right) \\ &= I_1^2 \exp j\{\omega_s\tau\}\left[1+\frac{1}{2}\alpha^2\cos(2\omega_r t)\right. \\ &\qquad + 2\alpha\cos(\omega_r t)\cos\left(\frac{\omega_r}{2}\tau\right) \\ &\qquad \left. + \frac{1}{2}\alpha^2\cos(\omega_r\tau)\right]. \end{aligned} \tag{26}$$

The FT of this expression with respect to τ yields the WD

$$\begin{aligned} W_{i_{\rm am}}(t,f) &= {\rm FT}_{\tau\to f}\left\{K_{i_{\rm am}}(t,\tau)\right\} \\ &= \left[1+\frac{1}{2}\alpha^2\cos(2\omega_r t)\right] I_1^2\,\delta(f-f_s) \\ &\quad + \alpha\cos(\omega_r t) I_1^2\,\delta\left(f-f_s\pm\frac{f_r}{2}\right) \\ &\quad + \frac{1}{4}\alpha^2 I_1^2\,\delta(f-f_s\pm f_r). \end{aligned} \tag{27}$$

An approximation for this expression can be obtained considering that the terms with α^2 can be neglected for small modulation indexes $\alpha \ll 1$

$$W_{i_{\rm am}}(t,f) \approx I_1^2\,\delta(f-f_s) + \alpha\cos(\omega_r t) I_1^2\,\delta\left(f-f_s\pm\frac{f_r}{2}\right). \tag{28}$$

The AM signature on the WD is therefore sidebands at $f_s \pm f_r/2$. The sidebands oscillate at shaft rotational frequency f_r; their amplitude is αI_1^2. It should be noted that the signature is similar to the PM signal but with the important difference that the upper and lower sideband oscillations have the same amplitudes for a given point in time, i.e., they are in phase.

D. Illustration With Synthesized Signals

In order to validate the preceding theoretical considerations, the periodogram and WD of the AM and PM signals are calculated numerically with synthesized signals. The signals are discrete versions of the continuous time signals in (3) and (12) with the following parameters: $I_{st} = I_{rt} = \sqrt{2}/2$, $I_1 = \sqrt{2}$, $\alpha = \beta = 0.1$, $\varphi_s = -\pi/8$, $f_s = 0.25$, and $f_c = f_r = 0.125$ normalized frequency. White zero-mean Gaussian noise is added with a signal-to-noise ratio of 50 dB. The signal length is $N = 512$ samples.

First, the periodogram of both signals is calculated (see Fig. 2). Both spectra show the fundamental component with sidebands at $f_s \pm f_r$. The higher order sidebands of the PM signal are buried in the noise floor so that both spectral signatures are identical.

The WD is often replaced in practical applications with the Pseudo Wigner Distribution (PWD). The PWD is a smoothed and windowed version of the WD, defined as follows [11]:

$${\rm PW}_x(t,f) = \int_{-\infty}^{+\infty} p(\tau) x\left(t+\frac{\tau}{2}\right) x^*\left(t-\frac{\tau}{2}\right) e^{-j2\pi f\tau} d\tau \tag{29}$$

where $p(\tau)$ is the smoothing window. In the following, a Hanning window of length $N/4$ is used. The time–frequency distributions are calculated using the Matlab time–frequency toolbox [14].

The PWD of the PM and AM stator current signals is displayed in Fig. 3. A constant frequency at $f_s = 0.25$ is visible in each case. Sidebands resulting from modulation appear at $f_s \pm f_r/2$ in both cases. The zoom on the interference structure shows that the sidebands are oscillating at f_r. According to the theory, the sidebands are phase shifted by approximately π in the PM case, whereas they are in phase with the AM signal.

IV. Experimental Results

A. Experimental Setup

Tests have been performed on an experimental setup (see Fig. 4) with a three-phase 400 V, 50 Hz 5.5 kW Leroy Somer induction motor (motor A). The motor has $p = 2$ pole pairs,

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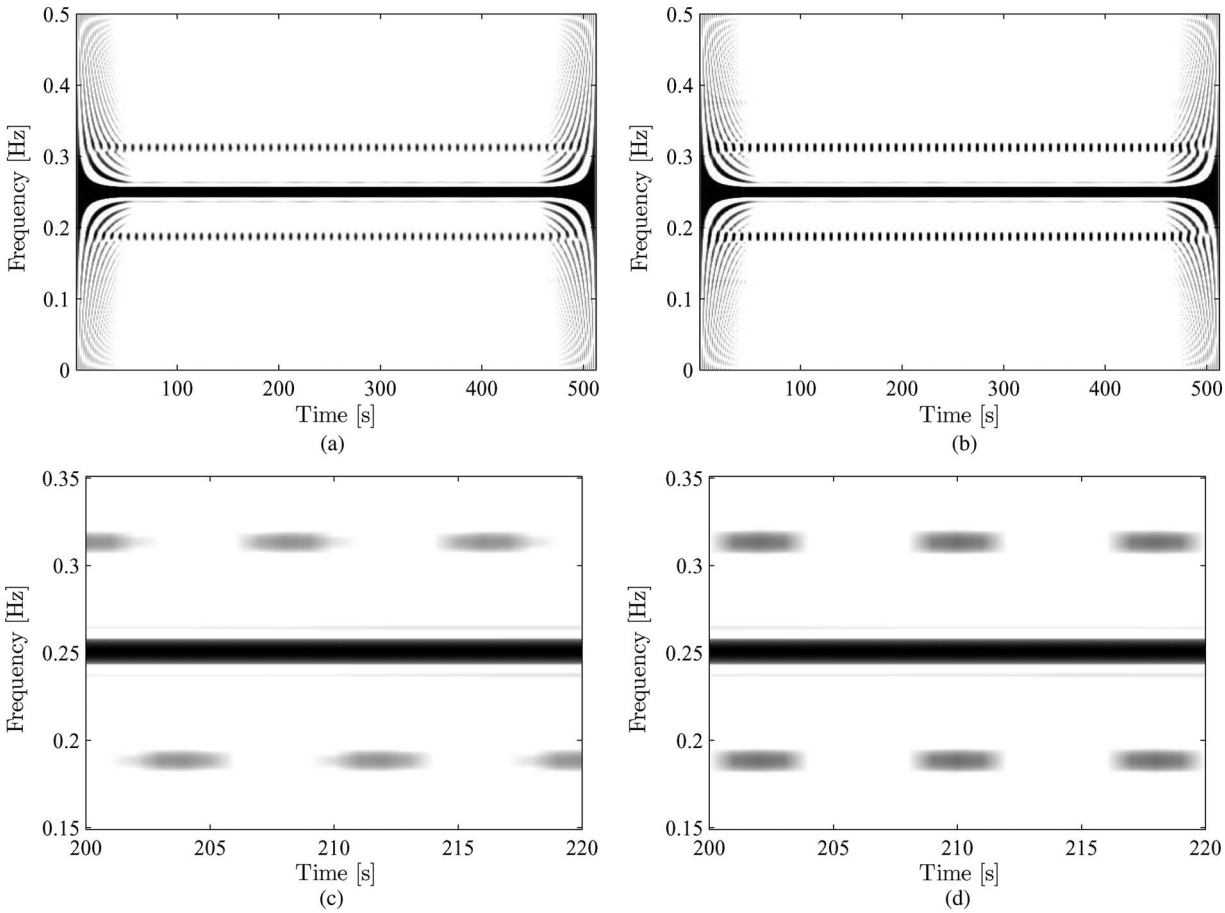


Fig. 3. Pseudo-WD of synthesized (a) PM and (b) AM signals with (c) and (d) zoom on interference structure.

and its nominal torque Γ_n is about 36 Nm. The machine is supplied by a standard industrial inverter operating in open-loop condition with a constant voltage to frequency ratio. The switching frequency is 3 kHz.

The load is a dc motor with separate constant excitation connected to a resistor through a dc/dc buck converter. A standard proportional–integral controller regulates the dc motor armature current. Thus, using an appropriate current reference signal, a constant torque with a small additional oscillating component can be introduced. The sinusoidal oscillation is provided through a voltage-controlled oscillator linked to a speed sensor. Since the produced load torque oscillations are not a realistic fault, load unbalance is also examined. Thereto, a mass is fixed on a disk mounted on the shaft. The torque oscillation produced by such a load balance is sinusoidal at the shaft rotational frequency. With the chosen mass and distance, the torque-oscillation amplitude is $\Gamma_c = 0.06$ Nm. If the motor bearings are healthy, then the additional centrifugal forces created by the mass will not lead to air gap eccentricity.

A second induction motor with identical parameters has been modified to introduce dynamic air gap eccentricity (motor B). Therefore, the bearings have been replaced with bearings having a larger inner diameter. Then, eccentrical fitting sleeves have been inserted between the shaft and the inner race. The obtained degree of dynamic eccentricity is approximately 40%.

Measured quantities in the experimental setup include the stator voltages and currents, torque, and shaft speed [see Fig. 4].

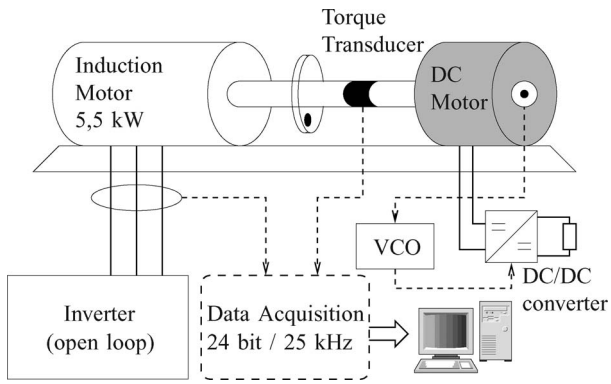


Fig. 4. Scheme of experimental setup.

The signals are simultaneously acquired through a 24 bit data acquisition board at 25 kHz sampling frequency. Further signal processing is done offline with Matlab.

B. Stator Current Spectrum Analysis

For illustration purposes, the stator current spectral signatures of a machine with dynamic eccentricity (motor B) are compared with an operation with load torque oscillations at frequency $f_c = f_r$ (motor A). The oscillating torque is produced through a regulation of the dc motor current. The stator current spectra show identical fault signatures around the fundamental

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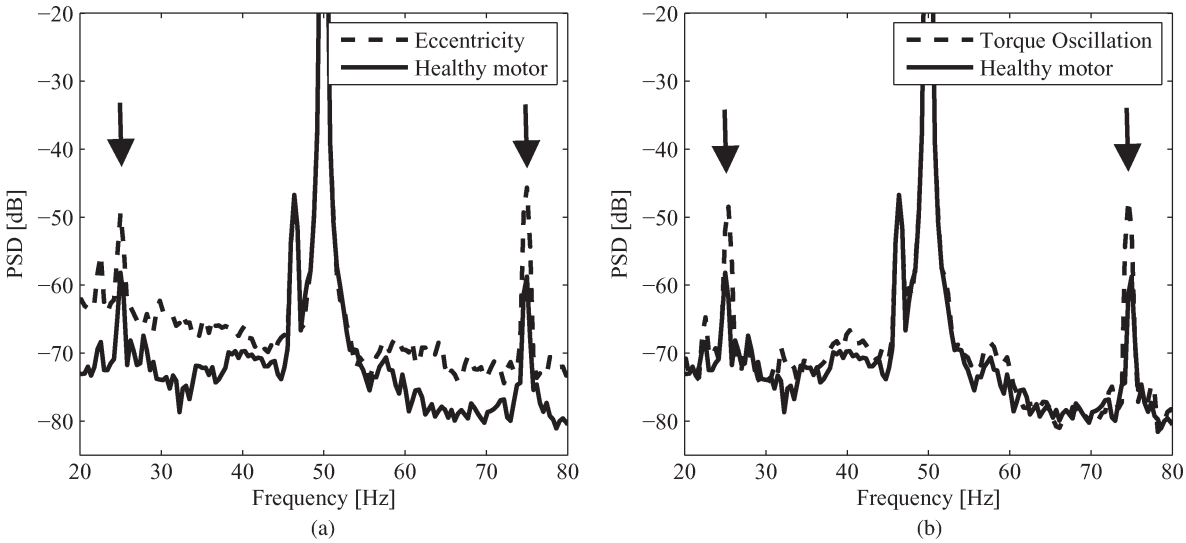


Fig. 5. Comparison of experimental motor stator current spectra: (a) 40% eccentricity (motor B) versus healthy machine (motor A). (b) 0.14 Nm load torque oscillation (motor A) versus healthy machine (motor A) at 10% average load.

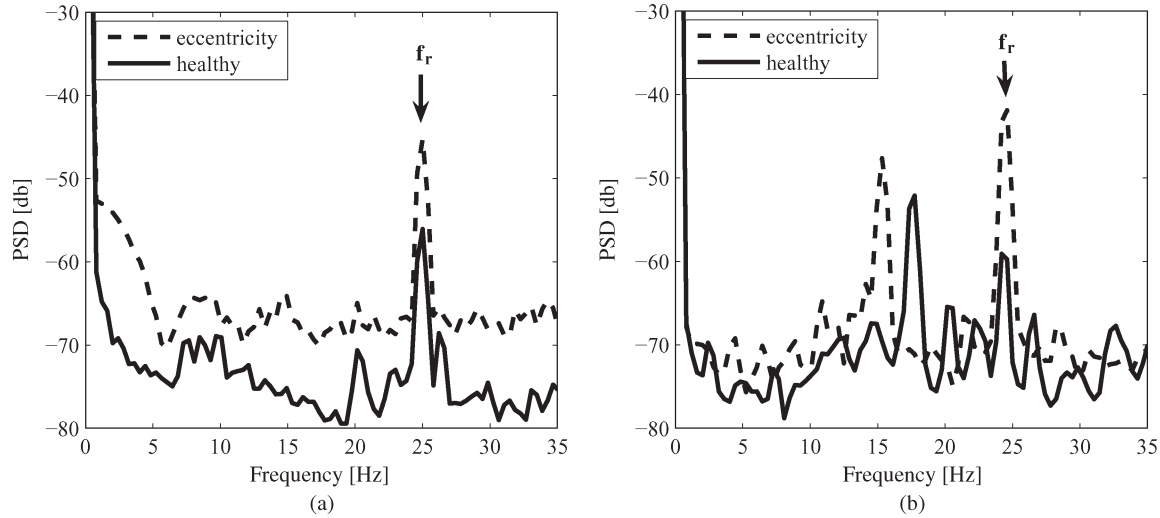


Fig. 6. Comparison of experimental torque spectra: 40% dynamic eccentricity (motor B) versus healthy machine (motor A) in different load conditions: (a) 10% load, (b) 80% load.

frequency i.e., an increasing amplitude of the peaks at $f_s \pm f_r$. This behavior is identical under different load conditions.

As an example, consider Fig. 5, where the current spectrum of a motor with 40% dynamic eccentricity is compared with an operation with load torque oscillations of amplitude $\Gamma_c = 0.14$ Nm. This corresponds to only 0.4% of the nominal torque. The healthy motor spectrum is also displayed, and the average load was 10% of nominal load during this test. It can clearly be recognized that a monitoring strategy based on the spectral components $f_s \pm f_r \approx 25$ and 75 Hz can be used efficiently for detection purposes. In both cases, these components show a considerable rise. However, this monitoring approach cannot distinguish the two fault types.

C. Torque Oscillations With Eccentricity

Torque measurements are performed to compare the oscillating components in healthy state and with dynamic eccentricity. The PSD of the measured torque is shown in Fig. 6 for two different load conditions. An oscillating torque component at shaft rotational frequency f_r exists already in the healthy case. With dynamic eccentricity, this component increases in amplitude by approximately 10 dB with 10% load and 16 dB with 80% load. These observations confirm the theoretical considerations in Section II-C.

D. Stator Current WD

Before the calculation of the stator current WD, the measured current signals are low pass filtered (cutoff frequency is 90 Hz) and downsampled to 200 Hz. This eliminates high-frequency harmonics that would generate interference terms on the WD. Then, the corresponding analytical signal is obtained through the Hilbert transform.

The stator current PWDs with dynamic eccentricity and with 0.14 Nm load torque oscillation are shown in Fig. 7 for 10%

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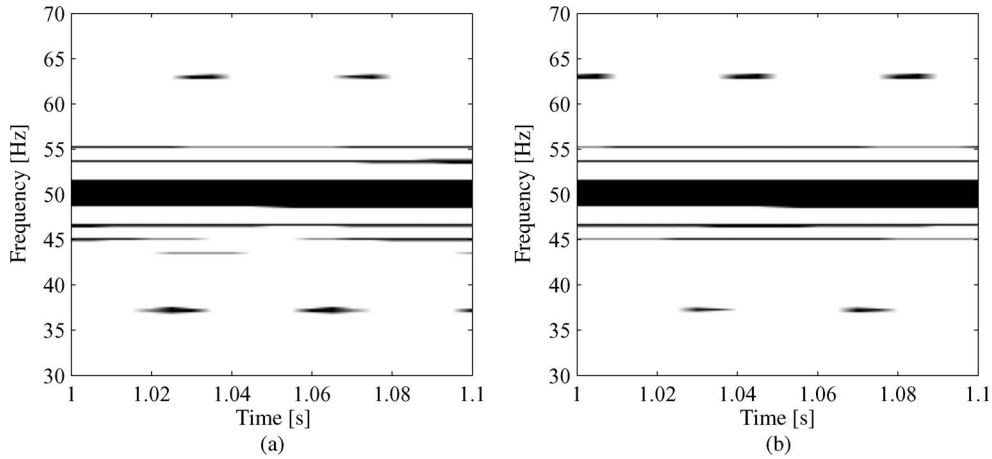


Fig. 7. Detail of stator current PWD with (a) 40% dynamic eccentricity (motor B) and (b) 0.14 Nm load torque oscillation (motor A) at small average load.

average load. The characteristic fault signature is visible in both cases at $f_s \pm f_r/2 = 37.5$ and 62.5 Hz. The phase shift between the upper and lower sideband seems closer to zero with eccentricity, whereas, with torque oscillations, it is closer to π. Anyhow, it is difficult to determine the exact value from the PWD.

In order to obtain a suitable fault indicator, the amplitude and phase shift of the two sidebands must be analyzed. This can be done by extracting two signals from the PWD that represent the upper and lower sideband. For a given time instant, the maximum of PWD yields the stator current fundamental supply frequency f_s. With the knowledge of the pole-pair number p, the maximum rotational frequency $f_{r,\max} = f_s/p$ at no load is calculated. The minimum rotational frequency $f_{r,\min}$ in steady state can be deduced from the nominal motor slip. The two sidebands are thus located in the intervals $I_l = [f_s - f_{r,\max}/2, f_s - f_{r,\min}/2]$ and $I_u = [f_s + f_{r,\min}/2, f_s + f_{r,\max}/2]$. The proposed algorithm determines the maximum absolute value in these intervals. The signed value from each interval is then taken to synthesize the lower and upper sideband signal. These signals oscillate theoretically at f_r and have amplitudes proportional to the PM or AM index, respectively.

The two signals corresponding to dynamic eccentricity and load torque oscillations are shown in Fig. 8. They are oscillating at the same frequency and their amplitudes are similar. The exact phase difference between the two signals is difficult to determine from the time waveforms because of the low sampling rate compared with the oscillation frequency f_r. As the stator current was downsampled to 200 Hz, the two sideband signals have only eight samples per period.

E. Fault Indicator

For calculation of a suitable fault indicator, the information about the sideband-signal amplitude and phase shift is extracted using the discrete FTs of the two signals. After computation of the FTs of the upper and lower sideband signals, the highest magnitudes in the frequency interval $[f_{r,\min}, f_{r,\max}]$ are determined. Both magnitudes from the upper and lower sideband signal are added. The sum is the first fault indicator A. Then, the phase shift φ between the two sideband signals is calculated by subtracting the corresponding phases from the FTs. All the phase information has been referred to $[0, \pi[$.

Since this monitoring method provides two quantities, amplitude A and phase shift φ, it can be graphically represented in a complex form $Ae^{j\varphi}$. The magnitude A contains information about the existence of a fault or its severity, whereas φ provides knowledge if eccentricity or torque oscillations prevail.

The obtained experimental results are displayed in Table I with the average indicator values for a healthy motor and different faults. The considered faults are 40% dynamic eccentricity, load unbalance, and load torque oscillations of different amplitude produced by variation of the dc motor current. For each load torque oscillation, the absolute amplitude and the relative amplitude in percent of the nominal torque Γ_n is indicated. The acquired current signals were analyzed using 21 data records, each of length 512 samples. The displayed magnitude a in the table is $1000 \cdot A$.

The fault indicator magnitude A shows a relevant increase comparing the healthy motor with the faulty cases. This can already be noticed with only 0.03 Nm torque oscillation which is less than 0.1% of the nominal torque. The mean phase shift φ is approximately 90° in the case of dynamic eccentricity, whereas with load torque oscillations, values between 125° and 165° are obtained. The angle values in the case of eccentricity are different from the theoretical value (around 0°) because of the previously discussed load torque oscillations in consequence of eccentricity. The obtained values signify that AM and PM can be found simultaneously in the case of eccentricity. Nevertheless, the two fault types can clearly be distinguished through the phase shift φ.

For illustration, the complex fault indicators $Ae^{j\varphi}$ for each data record are shown in Fig. 9 for the healthy case, eccentricity, and load torque oscillations of increasing amplitude. The average load level is 50%. The magnitude A of the healthy measurements is small, and, thus, they are located close to zero in the complex plane. The torque oscillations of increasing amplitude lead to an increasing indicator magnitude A. Their phase angle is close to π, which is a sign of PM. The fault-indicator values with dynamic eccentricity have a phase angle

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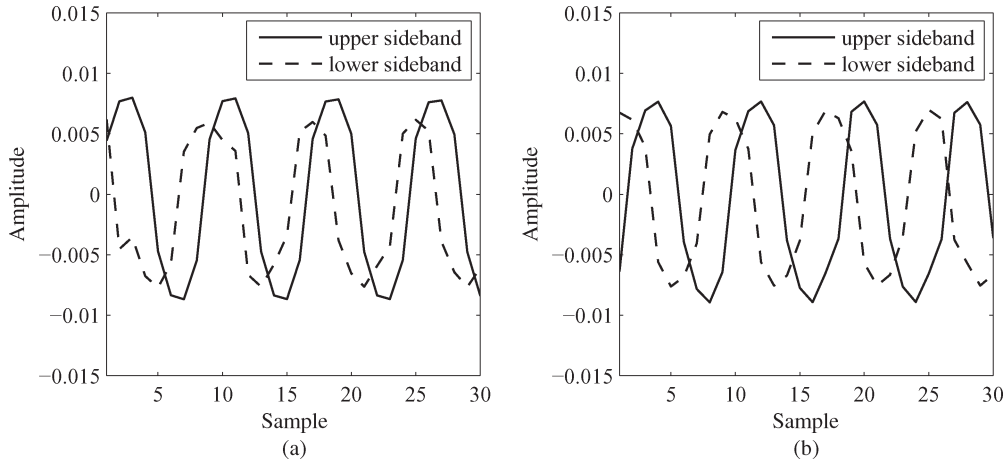


Fig. 8. Sideband signals extracted from stator current WD with (a) 40% dynamic eccentricity (motor B) and (b) 0.14 Nm load torque oscillation (motor A) at small average load. (a) Eccentricity. (b) Torque oscillation.

TABLE I
AVERAGE FAULT INDICATORS FOR HEALTHY MOTOR (MOTOR A), DYNAMIC ECCENTRICITY (MOTOR B), TORQUE OSCILLATIONS (MOTOR A), AND LOAD UNBALANCE (MOTOR A)

	10% load		50% load		80% load	
	a	φ	a	φ	a	φ
healthy	7.1	135°	8.3	158°	6.4	133°
40% dyn. eccentricity	23.6	82°	16.6	93°	15.9	86°
Torque oscillations Γ_c (% of Γ_n)						
0.03 Nm (0.08%)	10.8	126°	13.6	165°	10.3	137°
0.07 Nm (0.19%)	15.0	124°	19.2	165°	14.3	136°
0.11 Nm (0.31%)	19.3	129°	26.8	166°	20.0	136°
0.14 Nm (0.39%)	24.3	129°	36.7	163°	24.7	131°
0.18 Nm (0.50%)	30.1	123°	46.3	165°	30.0	135°
0.22 Nm (0.61%)	36.9	122°	55.9	164°	37.5	135°
unbalance 0.06 Nm	20.1	129°	25.1	158°	18.5	129°

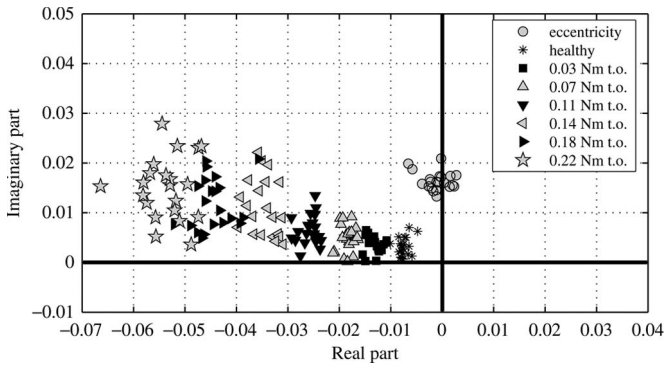


Fig. 9. Graphical representation of complex fault indicators with 50% average load: eccentricity (motor B), load torque oscillations (motor A), and healthy drive (motor A).

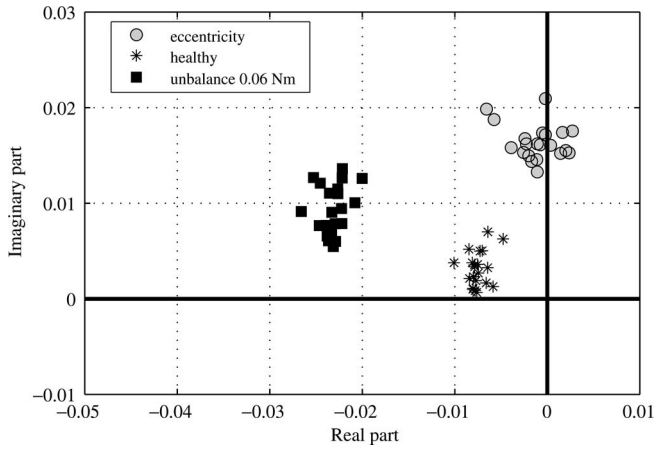


Fig. 10. Graphical representation of complex fault indicators with 50% average load: load unbalance (motor A), eccentricity (motor B), and healthy motor (motor B).

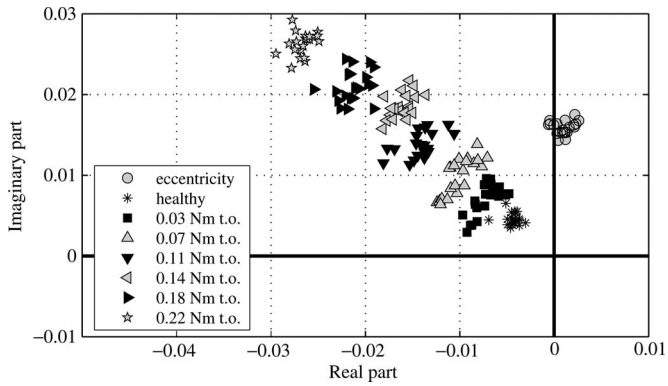


Fig. 11. Graphical representation of complex fault indicators with 80% average load: eccentricity (motor B), load torque oscillations (motor A), and healthy motor (motor A).

close to $\pi/2$, which points to the AM nature of the stator current signal.

The load unbalance is compared with the healthy case and eccentricity in Fig. 10. The result is similar to the previously displayed load torque oscillations with a phase angle close to π. This signifies that such an unbalance results mainly in stator current PM. Again, the distinction between the load unbalance, eccentricity, and healthy state is possible.

Additional results with 80% average load are displayed in Figs. 11 and 12. It can be noticed that the variance is smaller. The phase angles of the indicators with torque oscillations are approximately 135°, i.e., differing more from the theoretical value π than with 50% load. Nevertheless, the fault indicators with eccentricity can clearly be distinguished from the torque oscillations and the load unbalance. It can, therefore, be

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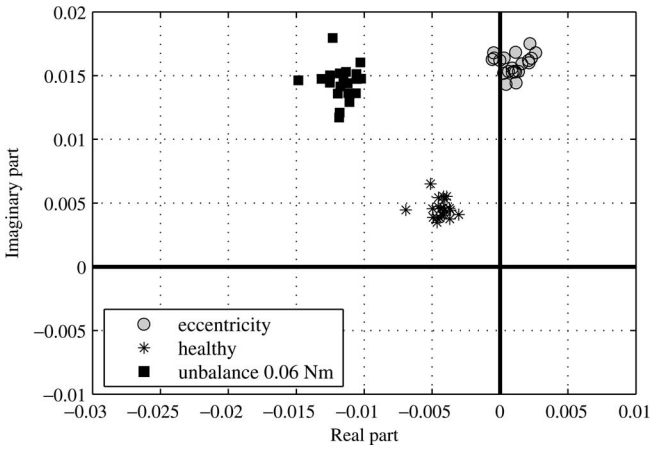


Fig. 12. Graphical representation of complex fault indicators with 80% average load: load unbalance (motor A), eccentricity (motor B), and healthy motor (motor A).

concluded that the proposed method allows a discrimination of eccentricity effects and small torque oscillations that cannot be realized with the classical spectrum estimation approach.

V. Conclusion

A novel method for stator current based diagnosis of mechanical faults in induction motor drives has been proposed in this paper. Through a theoretical approach, stator current models are derived that contain different modulations, AM and PM, caused by eccentricity and torque oscillations. Stator current time–frequency analysis using the WD allows the discrimination of AM and PM. Eccentricity effects and small torque oscillations can therefore be distinguished, which is impossible using classical current spectrum estimation. A fault indicator is proposed, and its effectiveness was demonstrated with experimental results under different load conditions and with different faults. The information about prevailing eccentricity or torque oscillations is useful for more accurate fault diagnosis in a monitoring system.

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Martin Blödt was born in Baden-Baden, Germany, in 1978. He received the Diploma degree in electrical engineering jointly from the University of Karlsruhe (TH), Karlsruhe, Germany, and from the Institut National Polytechnique de Grenoble, Grenoble, France, in 2003, and the Ph.D. degree from Institut National Polytechnique de Toulouse, Toulouse, France, in 2006.

In 2003, he joined the Laboratoire d'Electrotechnique et d'Electronique Industrielle (LEEI), Toulouse, where he conducted research on fault diagnosis in electrical machines and suitable signal-processing methods such as time–frequency signal processing and parameter estimation. Since 2007, he has been with the Mobility Division, Siemens AG, Erlangen, Germany.



Jérémi Regnier received the Ph.D. degree in electrical engineering from the Institut National Polytechnique de Toulouse (INP) de Toulouse, Toulouse, France, in 2003.

Since 2004, he has been an Assistant Professor in the Department of Electrical Engineering and Control Systems, ENSEEIHT, INP Toulouse. He is also a Researcher with the Laboratory of Plasma and Energy Conversion (Laboratoire Laplace), Toulouse. His research interests are modeling and simulation of faulty electrical machines and drives as well as the development of monitoring techniques using signal-processing methods.



Jean Faucher (M'06) received the Diploma degree in electrical engineering and the Ph.D. and Docteur ès Sciences degrees in switched reluctance machines from the Institut National Polytechnique (INP) de Toulouse, Toulouse, France, in 1967, 1969, and 1981, respectively.

Since 1985, he has been a Full University Professor at INP Toulouse, teaching courses on electrical and control engineering. He has been in charge of international relationships for INP Toulouse. He is also a Researcher with the Laboratory Laplace (Laboratory Plasma and Energy Conversion), Toulouse. His main fields of interest concern information processing and modeling of electrical machines and static converters for fault diagnosis, monitoring, and simulation.

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